

英語

問題Ⅰと問題Ⅱは別々の解答用紙に解答すること。

問題Ⅰ 次の英文を読んで次ページの以下の問い(1)～(3)に答えよ。

Many parents and day-care facilities nowadays expose tots to classical music in hopes of triggering the so-called “Mozart effect”—the sharpening of the brain that some classical music is said to bring about. Now the researchers who started the trend, Gordon Shaw and colleagues at the University of California, Irvine, have come up with evidence that piano lessons hike up children’s performance on a test of proportional math.

Six years ago, Shaw’s group found that listening to a Mozart two-piano sonata briefly raised college students’ spatial skills. They subsequently reported that in preschoolers, piano lessons gave a sustained boost to spatial skills.

In the latest study, Shaw compared three groups of second-graders: ①26 got piano instruction plus a math video game that trains players to mentally rotate shapes and to use them to learn ratios and fractions. Another 29 got computer-based English training plus the video game. A control group of 28 got no special training. After 4 months, the results were “dramatic,” ②the authors report in the current issue of Neurological Research.

The piano group scored 15% higher than the English group in a test of what they had learned in the computer game—and 27% higher on the questions devoted to proportional math. These gains were on top of the finding that the computer game alone boosted scores by 36% over the control group.

Shaw says the improvements suggest that spatial awareness and the need to think several steps ahead—both ③ in piano playing—reinforce latent neuronal patterns. “Music is just tapping into this internal neural structure that we’re born with,” he says. Piano lessons may well condition the brain just as muscle-building conditions an athlete, says Michael Merzenich, a neuroanatomist at the University of California, San Francisco. ④Music may be a “skill... more fundamental than language” for refining the ability of the brain to make spatial and temporal distinctions, he says.

(Science、1999年3月26日号、p.2007から引用)

注) tot: 幼児、hike up: 引き上げる、neuroanatomist: 神経解剖学者

英語

(1) 下線部①と④を和訳せよ。

(2) 下線部②で引用されている論文として最もふさわしいものは下記5編のうちどれか。番号で答えよ。

1. Johnson, J.K. *et al.*, "Enhancement of spatial-temporal reasoning after a Mozart listening condition in Alzheimer's disease: A case study," *Neurol. Res.* **20** (8): pp.666-672, 1998.
2. Kobayashi, H. *et al.*, "Hemodynamic and metabolic changes following cerebral revascularization in patients with cerebral occlusive diseases," *Neurol. Res.* **21** (2): pp.153-160, 1999.
3. Graziano, A.B. *et al.*, "Enhanced learning of proportional math through music training and spatial-temporal training," *Neurol. Res.* **21** (2): pp.139-152, 1999.
4. Ohaegbulam, S.C., "Geographical neurosurgery," *Neurol. Res.* **21** (2): pp.161-170, 1999.
5. Billiar, T. *et al.*, "Survival rates of malignant gliomas in Burgundy from 1990 to 1995," *Neurol. Res.* **21** (2): pp.171-174, 1999.

(3) 空白部③にあてはまる最適な単語は次のどれか。番号で答えよ。

1. compromised
2. required
3. ignored
4. measured
5. interested

問題Ⅱ 次の和文を英訳せよ。

外力無しにおいて物体は一定の速度を有する。もしも力が物体に働くならば、物体は力の大きさに比例した加速度を伴って力の方向に加速される。

注) 加速する : accelerate